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Name.....

Reg. No.....

**SIXTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION
MARCH 2025**

Mathematics

MTS6B12—CALCULUS OF MULTI VARIABLE

(Admissions Year—2019 Onwards)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer any number of questions.**Each question carries 2 marks.**Ceiling is 25.*

1. Find the domain of the function f defined by $f(x, y, z) = \sqrt{x + y - z} + xe^{yz}$.
2. Define limit of a function of two variables at a point.
3. Evaluate $\lim_{(x, y) \rightarrow (2, 4)} \sqrt[3]{\frac{8xy}{2x + y}}$.
4. Show that the function $u(x, y) = e^x \cos y$ is harmonic in the xy plane.
5. Let $f(x, y) = x^2 - 2xy$. Find the gradient of f at the point $(1, -2)$.
6. State second derivative test for a function of two variables.
7. Find the equation of the tangent plane to the graph of the function f defined by $f(x, y) = 4x^2 + y^2 + 2$ at the point $(1, 1)$.
8. Evaluate $\int_1^2 \int_0^1 3x^2 y \, dx \, dy$.
9. State Fubini's theorem for rectangular regions.

Turn over

10. Find the volume of the solid lying under the elliptic paraboloid $z = 8 - 2x^2 - y^2$ and above the rectangular region $R = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 2\}$.
11. Find the area of the part of the surface with equation $z = 2x + y^2$ that lies directly above the triangular region R in the xy -plane with vertices $(0, 0)$, $(1, 1)$ and $(0, 1)$.
12. A vector field $\mathbf{F}$ in $\mathbf{R}^2$ is defined by $\mathbf{F}(x, y) = -yi + xj$. Describe $\mathbf{F}$, and sketch a few vectors representing the vector field.
13. Find divergence of the vector field $\mathbf{F}(x, y, z) = yz \mathbf{i} + xz \mathbf{j} + xy \mathbf{k}$.
14. Find curl of the vector field $\mathbf{F} = x^2y^3 \mathbf{i} + xz^2 \mathbf{k}$.
15. State Green's theorem.

Section B

Answer any number of questions.

Each question carries 5 marks.

Ceiling is 35.

16. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $2x^2z - 3x^2 + yz - 8 = 0$.
17. Use chain rule to find $\frac{dw}{dt}$, where $w = 2x^3y^2z$, $x = t$, $y = \cos t$, $z = t \sin t$.
18. A heat-seeking object is located at the point $(2, 3)$ on a metal plate whose temperature at a point (x, y) is $T(x, y) = 30 - 8x^2 - 2y^2$. Find the path of the object if it moves continuously in the direction of maximum increase in temperature at each point.
19. Evaluate $\int_0^1 \int_y^1 \frac{\sin x}{x} dx dy$.

20. Evaluate $\iint_R y \, dA$, where R is the region in the first quadrant that is outside the circle $r = 2$, and inside the cardioid $r = 2(1 + \cos \theta)$.
21. Evaluate $\int_C 2x \, ds$, where C consists of the arc C_1 of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$ followed by the line segment C_2 from $(1, 1)$ to $(0, 0)$.
22. Evaluate $\int_C kz \, ds$, where k is a constant and C is the circular helix with parametric equations $x = \cos t$, $y = \sin t$ and $z = t$ where $0 \leq t \leq 2\pi$.
23. Let $\mathbf{F}(x, y) = 2xy \mathbf{i} + (1 + x^2 - y^2) \mathbf{j}$.
- (a) Show that $\mathbf{F}$ is conservative, and find a potential function f such that $\mathbf{F} = \nabla f$.
 - (b) If $\mathbf{F}$ is a force field, find the work done by $\mathbf{F}$ in moving a particle along any path from $(1, 0)$ to $(2, 3)$.

Section C

Answer any **two** questions.

Each question carries 10 marks.

Maximum 20 marks.

24. The production function of a certain country is given by $f(x, y) = 20x^{2/3}y^{1/3}$ billion dollars, when x billion dollars of labor and y billion dollars of capital are spent.
- (a) Compute $f_x(x, y)$ and $f_y(x, y)$.
 - (b) Compute $f_x(125, 27)$ and $f_y(125, 27)$ and interpret your result.
 - (c) Should the government encourage capital investment rather than investment in labor to increase the country's productivity?
25. Find the maximum and minimum values of the function $f(x, y) = x^2 - 2y$ subject to $x^2 + y^2 = 9$.

Turn over

26. Find the center of mass of the solid T of uniform density bounded by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = z$.
27. State divergence theorem. Let T be a solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = 3$, and let S be the surface of T . Calculate the outward flux of the vector field $\mathbf{F}(x, y, z) = xy^2 \mathbf{i} + yz^2 \mathbf{j} + zx^2 \mathbf{k}$ over S .

(2 × 10 = 20 marks)